Soft constraint: Cantera vs NN | Test IC 04

T_0 = 1317.5 K P_0 = 1.244 atm ϕ_0 = 1.984 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

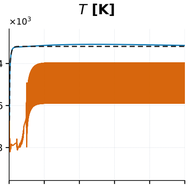
3 ANN10 + P-only DeepONet10

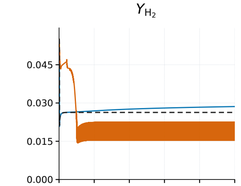
4 T/P DeepONet10

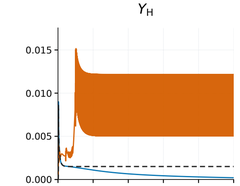
5 ANN10 + T/P DeepONet10

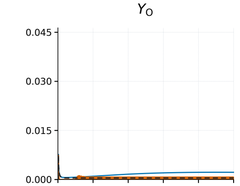
6 T/P/ϕ DeepONet10

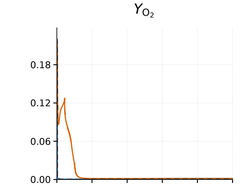
7 ANN10 + T/P/ϕ DeepONet10

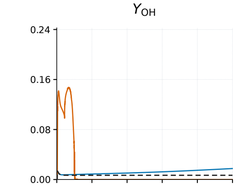


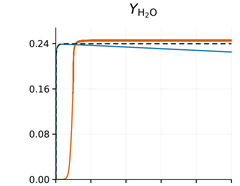


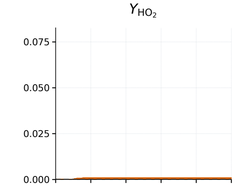


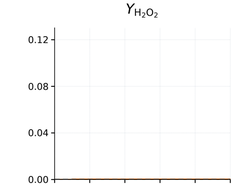


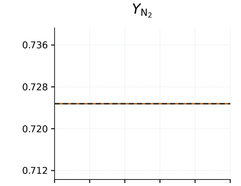


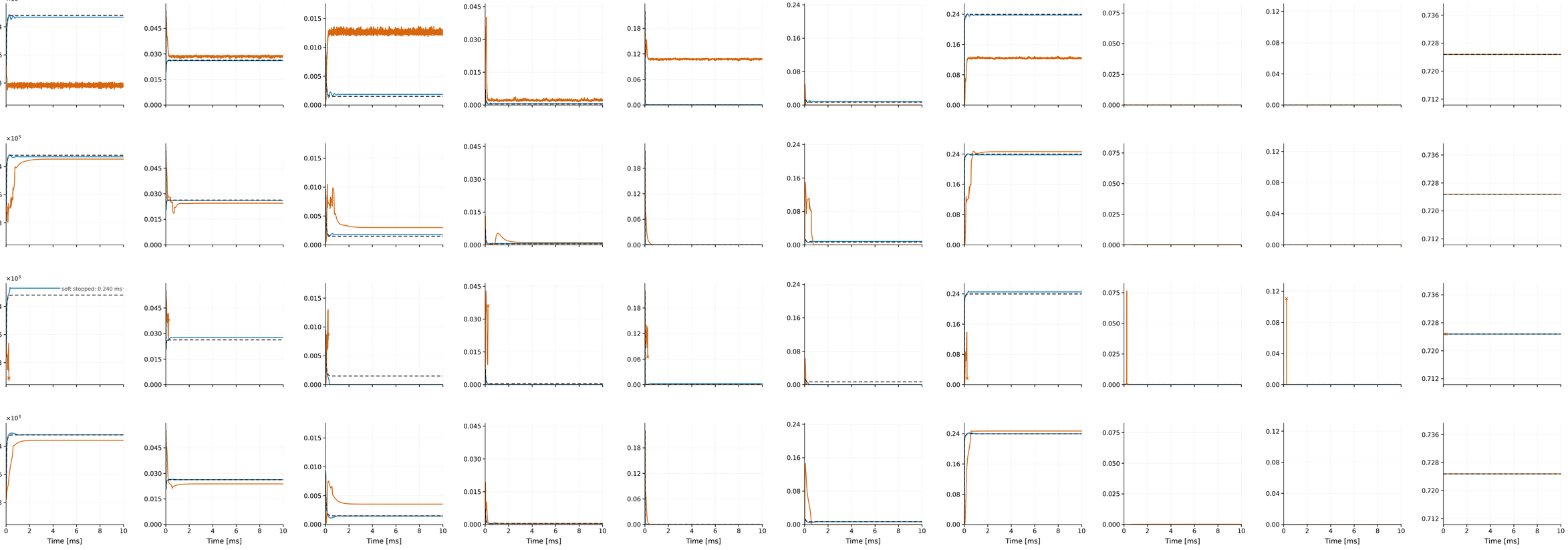












Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 04 | NRMSE (util.py)

T0 = 1317.5 K P0 = 1.244 atm phi0 = 1.984 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.0108	0.0583	0.593	2.33	0.0953	0.832	0.0354	0.149	0.418	0	Complete
1 ANN10	**soft10k**	0.346	0.354	4.88	0.747	2.76	5.37	0.288	170	740	0	Complete
2 P-only DeepONet10	**baseline**	8.74e-03	0.0113	0.165	0.217	0.0395	0.0155	2.22e-03	0.206	0.266	0	Complete
2 P-only DeepONet10	**soft10k**	0.29	0.486	1.13	3.72	2.47	8.33	0.704	5.9e+03	8.86e+05	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	9.87e-03	0.0137	0.196	0.258	0.0434	0.031	2.92e-03	0.155	0.2	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.262	0.385	2.17	8.88	4.02	6.24	0.44	4.9e+03	2.39e+05	0	Complete
4 T/P DeepONet10	**baseline**	0.0189	0.0104	0.227	0.392	0.0858	0.218	9.30e-03	0.336	0.28	0	Complete
4 T/P DeepONet10	**soft10k**	0.727	0.125	7.04	6.23	10.4	1.04	0.489	5.78	0.913	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	0.016	8.47e-03	0.186	0.267	0.0804	0.21	8.43e-03	0.233	0.181	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.137	0.11	1.45	2.28	0.847	3.46	0.134	59.7	588	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.0714	0.0485	0.943	0.811	0.196	0.973	0.0223	2.13	23.7	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.240 ms**
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	4.70e-03	3.41e-03	0.059	0.0529	0.0354	0.0625	2.65e-03	0.16	0.106	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.108	0.127	1.5	1.53	0.832	2.62	0.115	83.1	639	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 05

T_0 = 1049.0 K P_0 = 4.666 atm ϕ_0 = 0.674 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 05 | NRMSE (util.py)

T0 = 1049.0 K P0 = 4.666 atm phi0 = 0.674 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.299	0.773	1.07	1.02	0.614	11.4	0.509	1.03	1.01	0	Complete
1 ANN10	**soft10k**	0.318	0.922	10.6	5.47	0.659	6.66	0.585	121	6.62	0	Complete
2 P-only DeepONet10	**baseline**	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
2 P-only DeepONet10	**soft10k**	0.31	0.844	1	1	0.909	1	0.669	507	872	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.301	0.912	10.3	15	0.687	6.74	0.576	96.3	13.6	0	Complete
4 T/P DeepONet10	**baseline**	0.35	0.942	2.92	0.887	0.604	1.32	0.626	0.989	0.997	0	Complete
4 T/P DeepONet10	**soft10k**	0.36	0.909	7.05	1.02	0.56	1.25	0.627	1.03	1	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	0.34	0.913	2.09	0.728	0.586	1.34	0.606	0.921	0.996	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.342	0.968	11.8	9.17	0.631	1.8	0.656	42.5	1.31	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.565	1.48	1	1	0.929	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.388	0.976	8.65	1.03	0.598	1.29	0.675	3.94	3.85	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.565	1.48	1	1	0.929	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.354	0.952	6.19	5.75	0.621	1.69	0.651	74.1	3.15	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 07

T_0 = 930.6 K P_0 = 3.722 atm ϕ_0 = 0.754 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

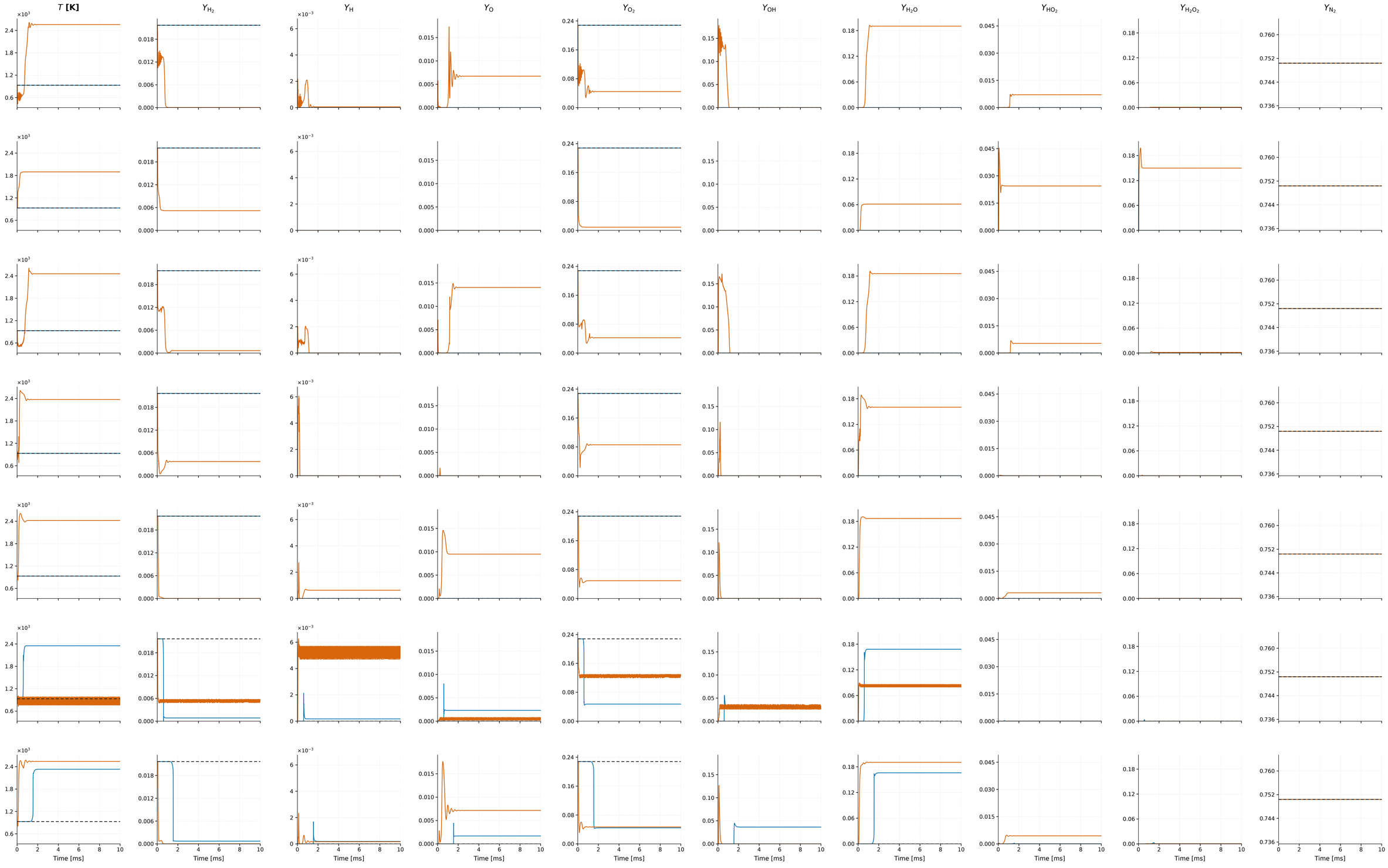
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 07 | NRMSE (util.py)

T0 = 930.6 K P0 = 3.722 atm phi0 = 0.754 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	2.26e-06	1.72e-06	1	360	2.48e-06	1	1	0.531	0.464	0	Complete
1 ANN10	**soft10k**	1.66	0.968	1.05e+08	8.70e+08	0.791	8.82e+09	1.09e+06	1.38e+04	2.24e+03	0	Complete
2 P-only DeepONet10	**baseline**	2.84e-06	2.61e-06	1	0.68	4.31e-06	0.986	0.999	0.994	1	0	Complete
2 P-only DeepONet10	**soft10k**	1.03	0.756	1.40e+05	1.48e+05	0.958	1.20e+05	3.63e+05	5.08e+04	3.81e+05	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	2.84e-06	2.61e-06	1	0.623	4.31e-06	0.987	0.999	0.992	1	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	1.56	0.942	1.11e+08	1.77e+09	0.802	9.78e+09	1.06e+06	1.04e+04	4.12e+03	0	Complete
4 T/P DeepONet10	**baseline**	2.85e-06	2.61e-06	29.5	0.994	4.31e-06	1.56	1	0.993	1	0	Complete
4 T/P DeepONet10	**soft10k**	1.54	0.832	1.76e+08	1.19e+07	0.628	2.08e+09	9.60e+05	70.1	285	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	2.86e-06	2.49e-06	105	0.972	4.01e-06	14.1	0.99	0.861	0.984	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	1.59	0.993	1.77e+08	1.28e+09	0.781	2.22e+09	1.11e+06	6.2e+03	663	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	1.48	0.931	5.41e+07	3.02e+08	0.768	6.67e+09	9.77e+05	107	599	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.0976	0.755	1.48e+09	6.75e+07	0.453	6.92e+09	4.97e+05	2.18	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	1.38	0.888	5.23e+07	2.11e+08	0.74	7.44e+09	9.16e+05	119	900	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	1.71	0.993	6.20e+07	9.94e+08	0.791	2.27e+09	1.13e+06	8.85e+03	1.14e+03	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 18

T_0 = 1069.1 K P_0 = 2.170 atm ϕ_0 = 1.544 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

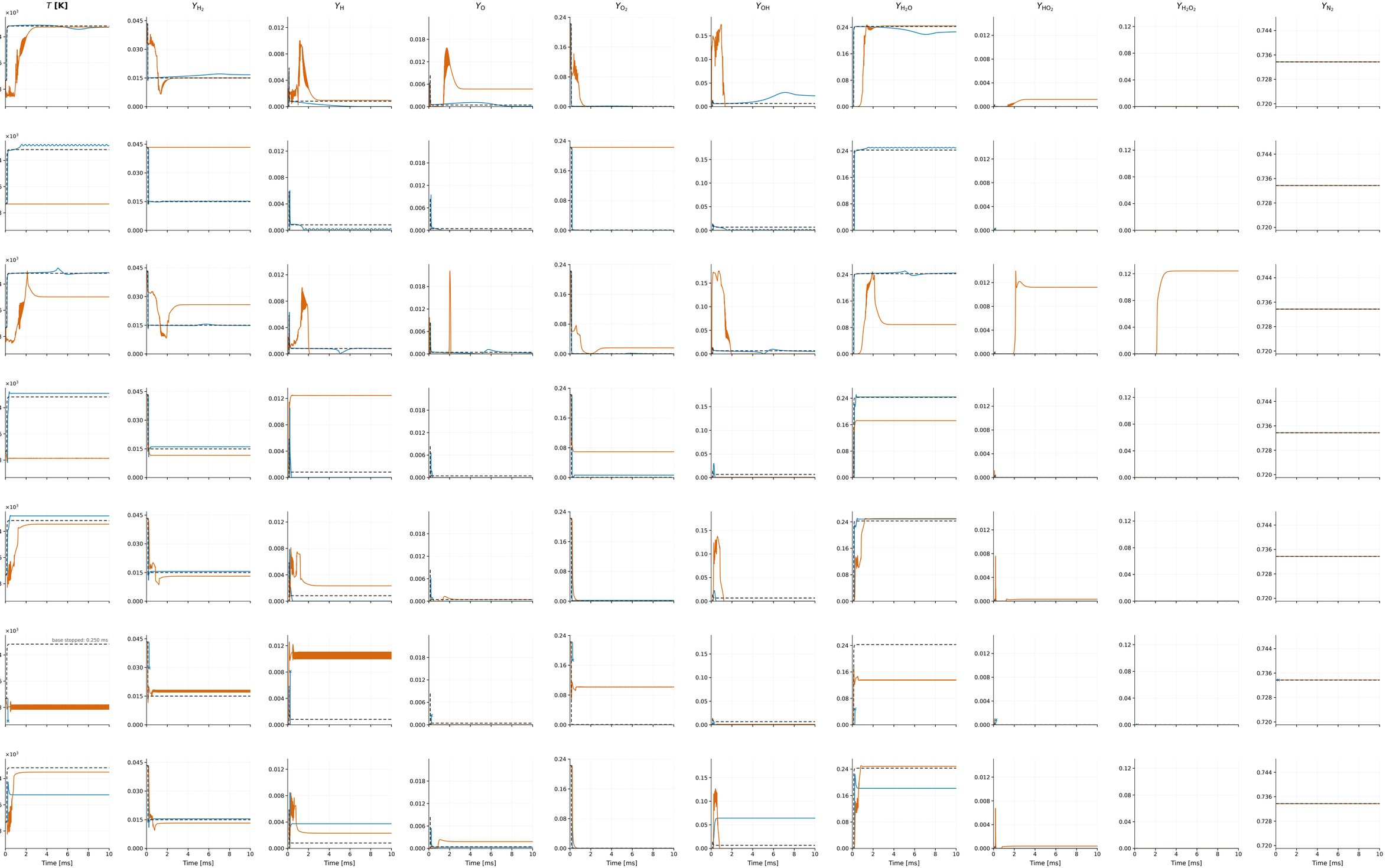
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 18 | NRMSE (util.py)

T0 = 1069.1 K P0 = 2.170 atm phi0 = 1.544 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.0255	0.115	0.793	0.959	0.336	1.94	0.0671	1.02	1.19	0	Complete
1 ANN10	**soft10k**	0.249	0.34	2.24	10	1.06	6.81	0.295	75.5	106	0	Complete
2 P-only DeepONet10	**baseline**	0.067	0.161	0.806	1.05	0.768	0.82	0.0892	1.54	2.36	0	Complete
2 P-only DeepONet10	**soft10k**	0.608	1.8	1	1	8.84	1	1	1	1	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	0.029	0.0992	0.314	0.756	0.462	0.297	0.051	1.31	1.86	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.351	0.681	2.4	4.07	0.985	8.15	0.637	706	9.43e+04	0	Complete
4 T/P DeepONet10	**baseline**	0.0652	0.169	1.06	1.06	0.778	1.03	0.0828	1.98	5.67	0	Complete
4 T/P DeepONet10	**soft10k**	0.687	0.221	13.7	0.996	2.76	0.914	0.293	2.67	0.999	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	0.0681	0.149	1.03	1.02	0.693	1.02	0.0796	1.62	2.2	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.172	0.24	2.57	0.648	0.895	4.41	0.175	24.9	70.6	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 0.250 ms**
6 T/P/ϕ DeepONet10	**soft10k**	0.703	0.185	11.5	0.954	4.01	0.874	0.441	2.19	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.304	0.175	3.44	0.629	0.833	8.52	0.265	2.04	5.77	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.143	0.214	2.3	2.77	0.856	3.11	0.142	28.9	47.4	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 32

T_0 = 1140.7 K P_0 = 3.354 atm ϕ_0 = 1.463 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

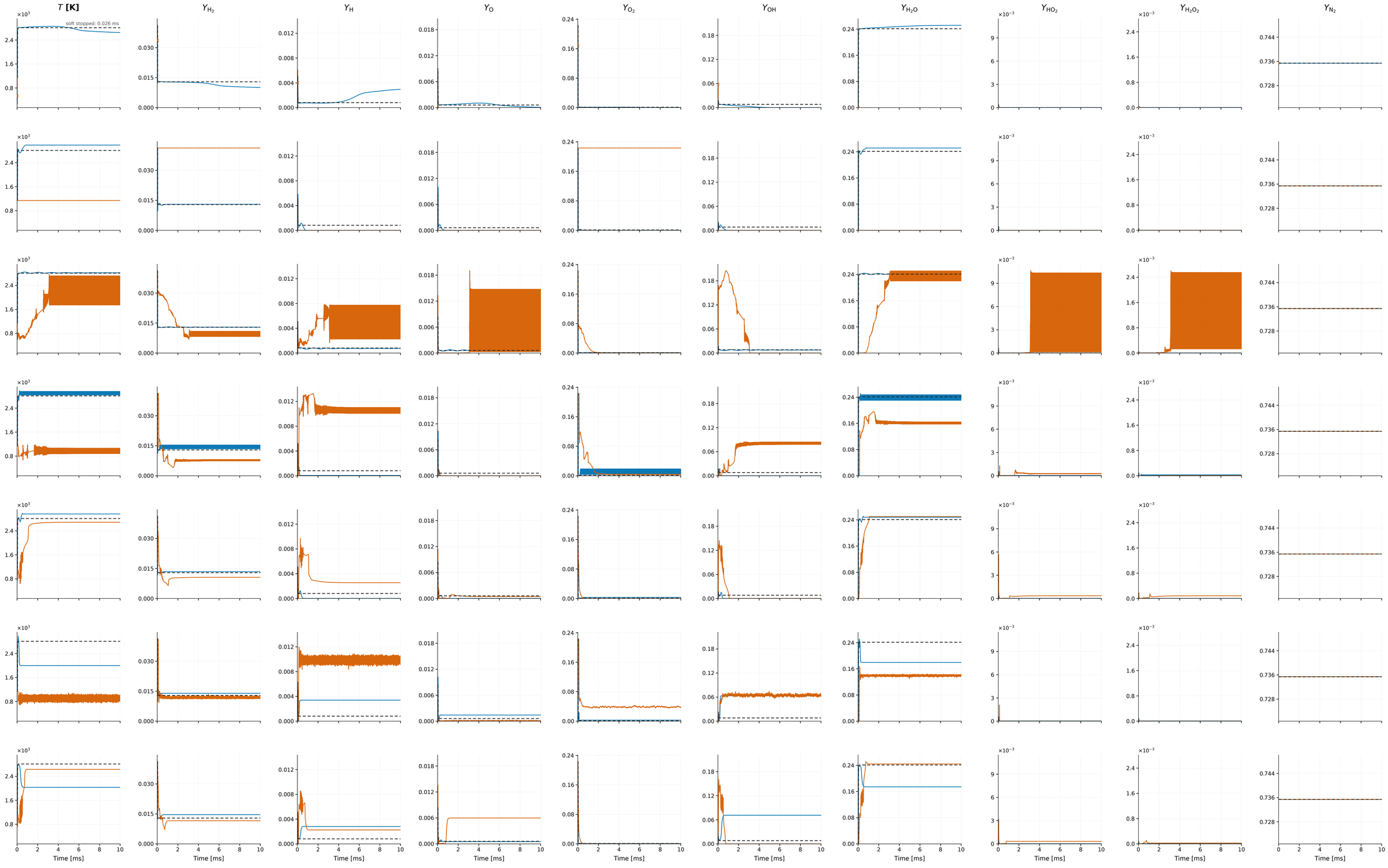
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 32 | NRMSE (util.py)

T0 = 1140.7 K P0 = 3.354 atm phi0 = 1.463 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

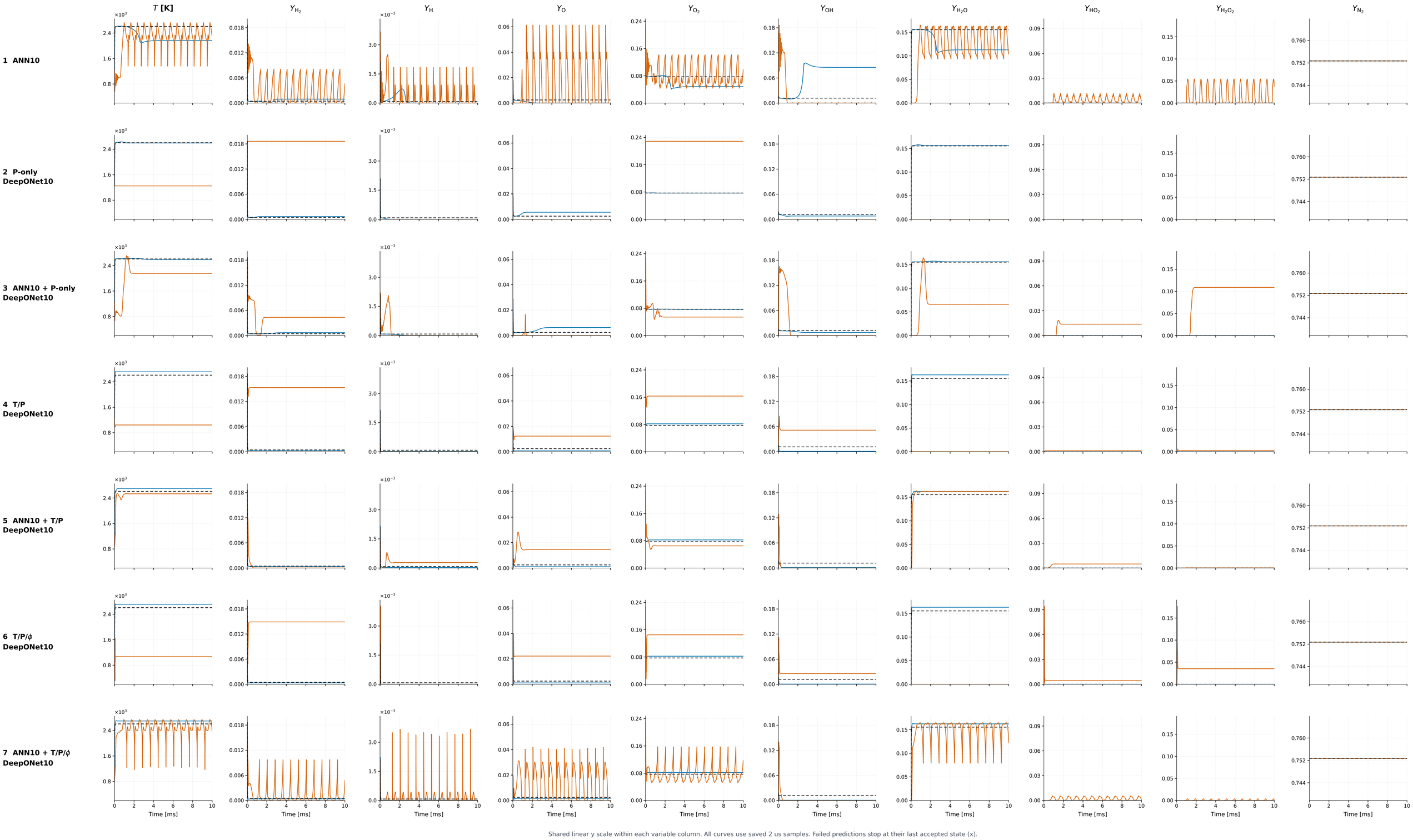
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.0312	0.126	1.48	0.511	0.0533	0.83	0.0317	0.118	1.03	0	Complete
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.026 ms**
2 P-only DeepONet10	**baseline**	0.0629	0.063	0.97	0.99	0.447	0.972	0.0492	1.24	1.51	0	Complete
2 P-only DeepONet10	**soft10k**	0.594	2.15	1	1	16.2	1	1	1.05	2.87	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	8.62e-03	0.0401	0.15	0.275	0.276	0.0993	0.0162	0.778	1	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.395	0.466	5.64	13.2	1.17	9.18	0.386	543	1.75e+03	0	Complete
4 T/P DeepONet10	**baseline**	0.04	0.127	0.968	1	0.844	0.993	0.0346	2.01	13	0	Complete
4 T/P DeepONet10	**soft10k**	0.657	0.458	12.2	0.996	2.43	8.18	0.341	24.4	1.53	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	0.0538	0.0417	0.965	0.912	0.184	0.98	0.0292	0.891	0.997	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.162	0.248	3.14	0.602	0.721	3.12	0.142	31.1	91	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.285	0.0903	3.13	1.31	0.345	6.75	0.254	2.61	7.38	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.679	0.207	11.1	0.879	3.13	6.86	0.428	5.83	1.02	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.272	0.13	2.39	0.126	0.0193	7.65	0.272	1.92	9.92	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.154	0.173	2.44	8.16	0.604	3.25	0.134	29.5	33.4	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 39

T_0 = 1250.1 K P_0 = 2.297 atm ϕ_0 = 0.646 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 39 | NRMSE (util.py)

T0 = 1250.1 K P0 = 2.297 atm phi0 = 0.646 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.15	0.476	2.07	0.897	0.329	5.49	0.241	0.78	0.837	0	Complete
1 ANN10	**soft10k**	0.216	4.61	6	8.07	0.391	2.82	0.329	286	2.53e+04	0	Complete
2 P-only DeepONet10	**baseline**	3.93e-03	0.236	0.758	1.17	6.65e-03	0.321	6.61e-03	0.331	0.24	0	Complete
2 P-only DeepONet10	**soft10k**	0.52	20.5	1	1	1.95	1	1	0.908	0.791	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	5.60e-03	0.233	0.703	1.3	7.30e-03	0.33	6.99e-03	0.292	0.178	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.252	4.86	4.23	1.06	0.29	3.59	0.601	700	1.01e+05	0	Complete
4 T/P DeepONet10	**baseline**	0.0406	0.212	0.694	0.675	0.0647	0.897	0.0488	0.769	0.794	0	Complete
4 T/P DeepONet10	**soft10k**	0.6	16.8	0.946	3.92	1.11	3.4	0.999	72.2	3.24e+03	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	0.0362	0.13	0.568	0.562	0.0688	0.891	0.0437	0.833	0.881	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.0613	1.32	2.39	4.95	0.172	1.21	0.0981	253	372	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.0408	0.197	0.787	0.562	0.0679	0.955	0.0499	0.634	0.567	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.591	16.2	1.53	7.82	0.873	1.23	1	299	3.62e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.0351	0.121	0.697	0.309	0.0652	0.934	0.0446	0.686	0.692	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.116	2.76	4.81	6.51	0.303	1.37	0.156	163	1.43e+03	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 44

T_0 = 1031.1 K P_0 = 2.272 atm ϕ_0 = 0.770 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

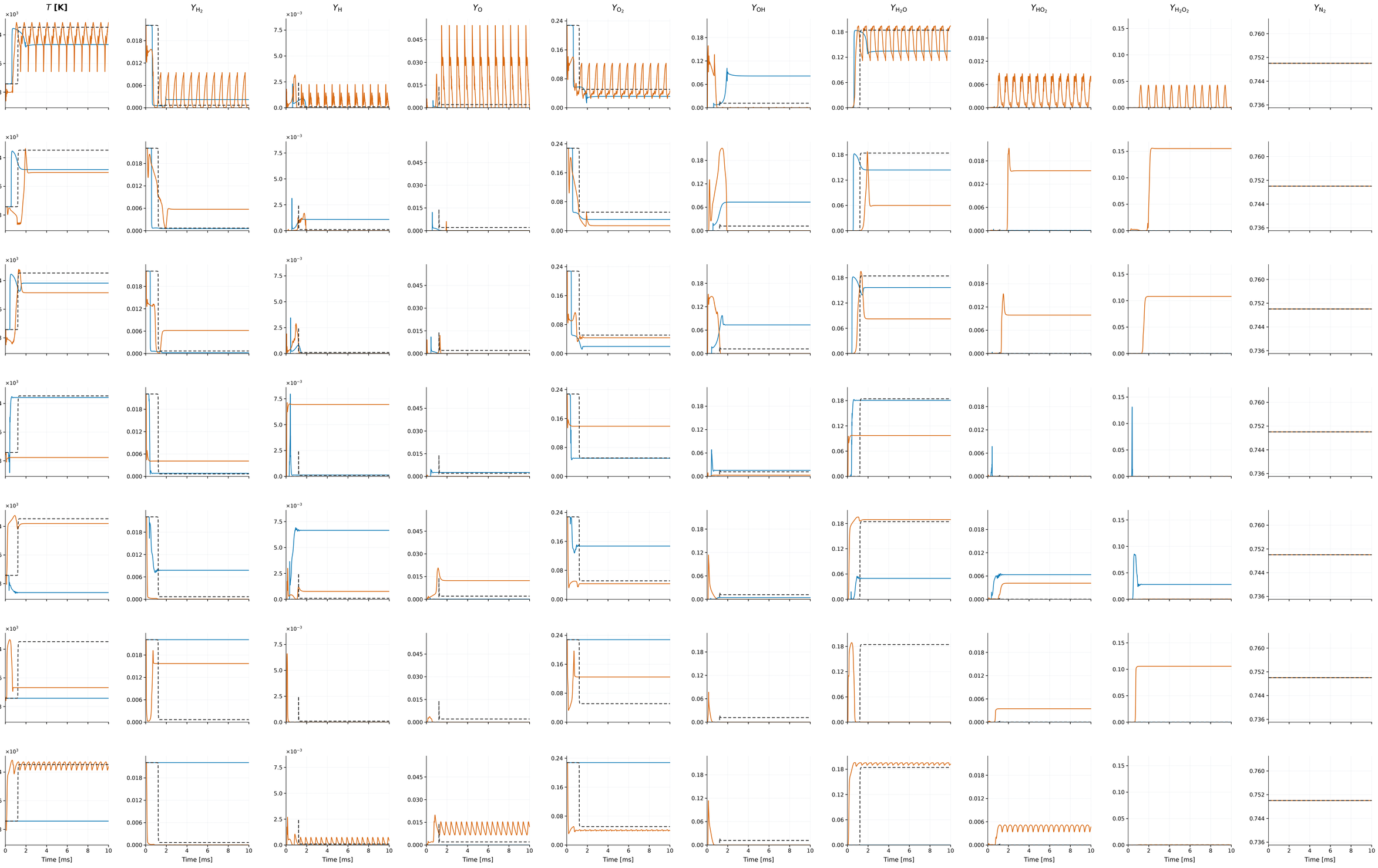
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 44 | NRMSE (util.py)

T0 = 1031.1 K P0 = 2.272 atm phi0 = 0.770 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.231	0.688	2.26	0.999	0.481	5.74	0.363	1.11	1.23	0	Complete
1 ANN10	**soft10k**	0.191	0.812	6.86	8.51	0.613	2.9	0.287	163	2.13e+03	0	Complete
2 P-only DeepONet10	**baseline**	0.252	0.701	8.52	0.997	0.528	5.19	0.335	1.82	1.38	0	Complete
2 P-only DeepONet10	**soft10k**	0.291	0.676	2.81	1	0.519	5.31	0.681	504	1.70e+04	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	0.194	0.773	1.77	1.02	0.628	5.32	0.317	1.03	0.986	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.214	0.867	4.32	1.02	0.538	3.95	0.556	337	1.22e+04	0	Complete
4 T/P DeepONet10	**baseline**	0.169	0.784	4.31	0.488	0.547	0.67	0.288	10.8	819	0	Complete
4 T/P DeepONet10	**soft10k**	0.652	0.881	62.2	0.999	0.943	0.744	0.509	0.93	1	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	0.782	0.972	57.9	0.965	1	0.638	0.733	221	3.74e+03	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.203	0.927	6.14	5.06	0.655	1.4	0.347	139	39.9	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.599	2.6	1	1	1.79	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.505	1.94	5.23	1.04	0.896	1.14	1.03	120	1.25e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.199	0.937	3.34	4.49	0.672	1.43	0.353	161	45.7	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 45

T_0 = 1181.9 K P_0 = 4.842 atm ϕ_0 = 1.099 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

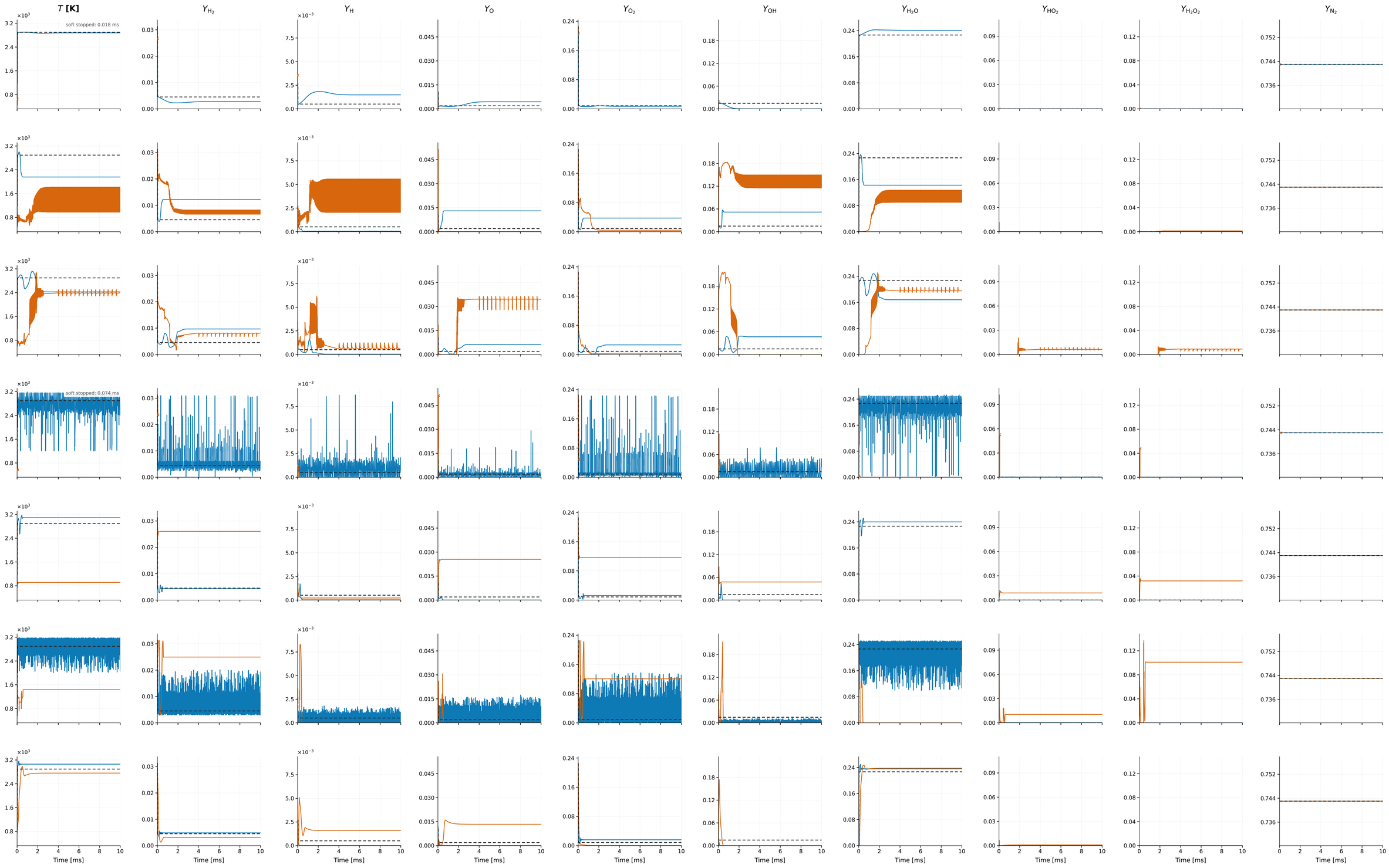
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 45 | NRMSE (util.py)

T0 = 1181.9 K P0 = 4.842 atm phi0 = 1.099 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	8.22e-03	0.379	1.98	1.07	0.214	0.942	0.0629	1.61	5.76	0	Complete
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
2 P-only DeepONet10	**baseline**	0.25	1.59	0.904	5.59	2.08	2.48	0.364	2	7.4	0	Complete
2 P-only DeepONet10	**soft10k**	0.574	1.23	6.93	1.23	1.44	8.57	0.609	16.1	583	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	0.148	0.982	0.896	1.99	1.19	2.01	0.236	2.46	12.9	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.311	1.05	2.13	15	0.696	4.61	0.35	399	5.68e+03	0	Complete
4 T/P DeepONet10	**baseline**	0.137	1.06	1.85	0.835	2.96	1.05	0.199	4.01	5.37	0	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.074 ms**
5 ANN10 + T/P DeepONet10	**baseline**	0.0687	0.168	0.958	0.974	0.537	0.996	0.0649	1.19	0.682	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.684	4.53	0.583	12.1	7.96	2.26	1	626	2.30e+04	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.0835	0.76	0.851	1.55	2.22	0.889	0.134	2.86	5.55	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.511	4.29	2.03	1.36	8.35	1.52	0.993	750	7.11e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.0594	0.181	0.981	0.969	0.723	0.997	0.0499	0.892	0.616	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.0934	0.457	2.4	5.81	0.839	1.42	0.108	63.1	21	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 53

T_0 = 1257.8 K P_0 = 2.103 atm ϕ_0 = 1.799 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 53 | NRMSE (util.py)

T0 = 1257.8 K P0 = 2.103 atm phi0 = 1.799 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.0249	0.105	0.84	0.642	0.0369	3.25	0.0957	0.118	0.566	0	Complete
1 ANN10	**soft10k**	0.351	0.399	5.59	0.977	2.4	7.79	0.329	79.8	397	0	Complete
2 P-only DeepONet10	**baseline**	0.0551	0.0218	0.754	0.845	0.0818	0.868	0.03	0.247	0.641	0	Complete
2 P-only DeepONet10	**soft10k**	0.545	1.32	1	1	23.3	1	1	0.908	0.929	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	2.57e-03	2.74e-03	0.0432	0.0484	0.0322	0.0285	1.59e-03	0.131	0.184	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.334	0.512	2.19	6.66	2.08	8.27	0.653	3.08e+03	5.41e+05	0	Complete
4 T/P DeepONet10	**baseline**	0.0311	0.0186	0.439	0.467	0.196	0.463	0.0176	0.488	0.557	0	Complete
4 T/P DeepONet10	**soft10k**	0.628	1.02	0.993	26.2	11.1	14.1	1	369	1.70e+04	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	0.0172	0.0109	0.236	0.198	0.13	0.287	0.0107	0.341	0.331	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.185	0.164	2.43	0.491	1.03	5.08	0.175	51.1	427	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.149	0.0297	1.67	1.47	0.293	3.28	0.106	1.41	6.57	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.715	0.135	9.05	4.13	10.8	2.04	0.463	14.7	3.79	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.0109	0.0127	0.15	0.173	0.181	0.196	9.52e-03	0.429	0.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.139	0.145	1.99	0.798	0.952	3.63	0.135	68	375	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 57

T_0 = 1412.0 K P_0 = 4.083 atm ϕ_0 = 0.677 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

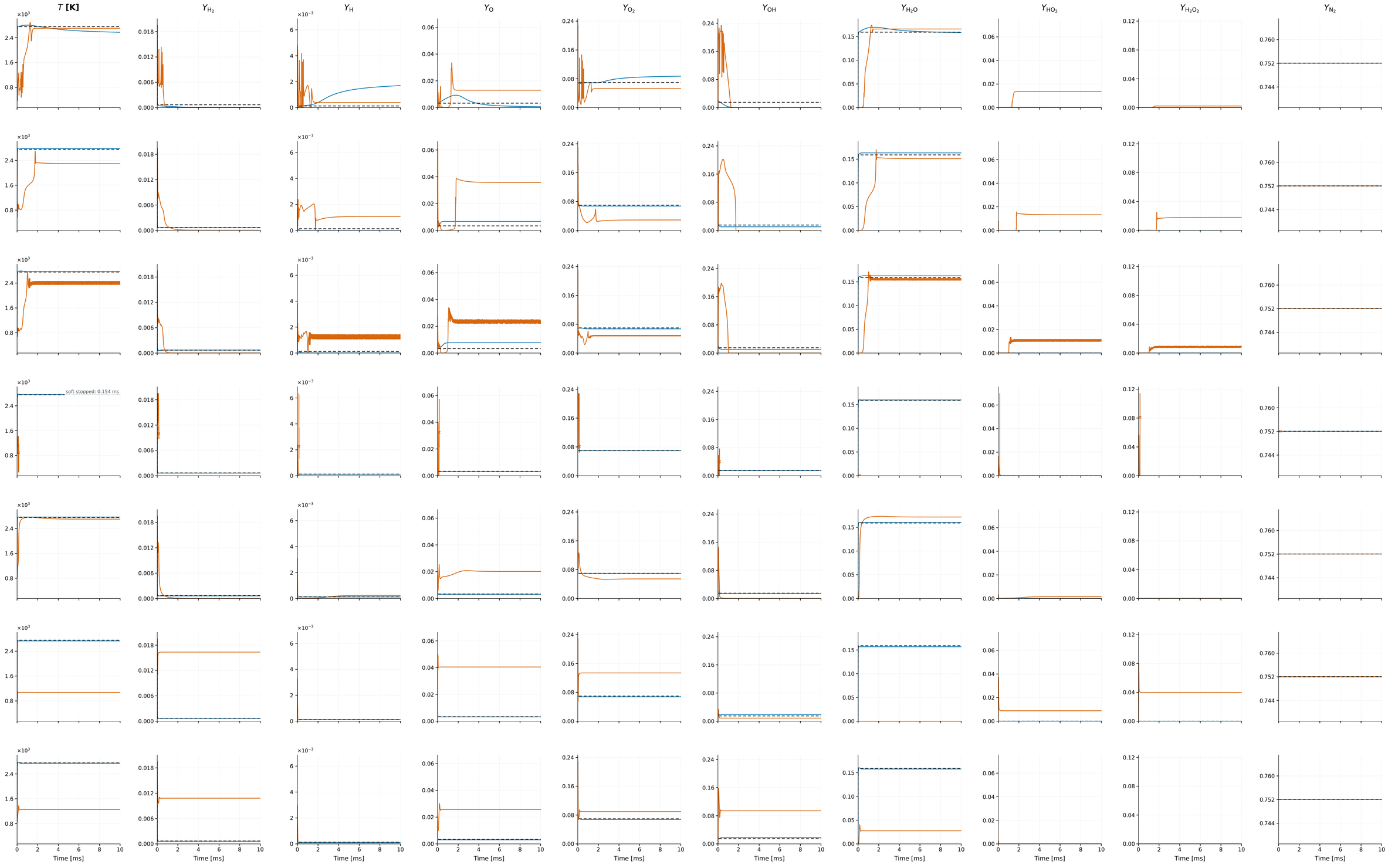
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 57 | NRMSE (util.py)

T0 = 1412.0 K P0 = 4.083 atm phi0 = 0.677 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

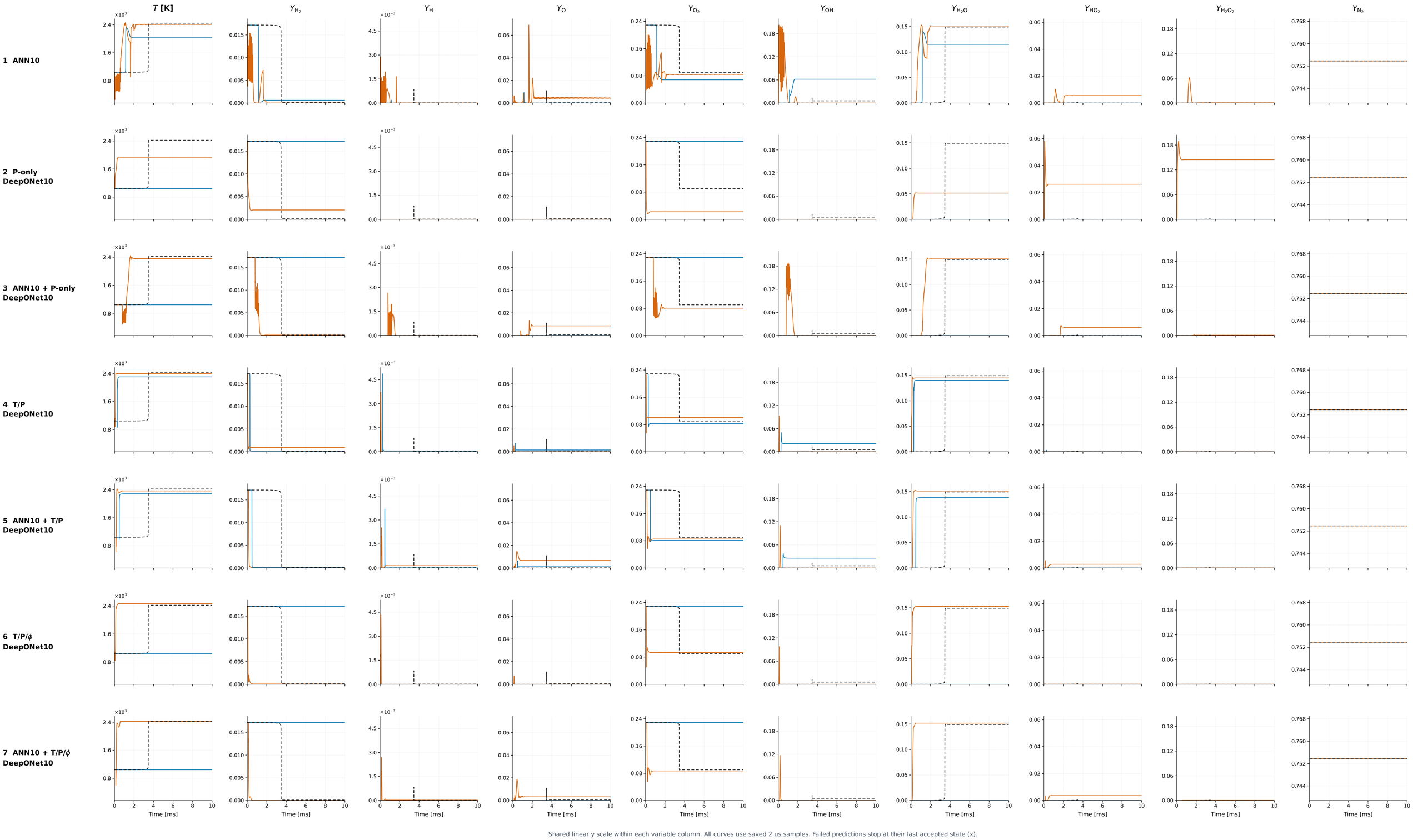
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.0428	0.666	8.62	0.853	0.176	0.956	0.0319	1.51	4.49	0	Complete
1 ANN10	**soft10k**	0.178	2.17	4.29	2.86	0.305	3.41	0.261	470	1.22e+03	0	Complete
2 P-only DeepONet10	**baseline**	0.0101	0.104	0.933	0.986	0.0348	0.33	0.0267	0.132	1.47	0	Complete
2 P-only DeepONet10	**soft10k**	0.266	2.27	8.16	8.9	0.588	4.05	0.307	449	9.7e+03	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	5.19e-03	0.0455	0.906	1.27	0.0366	0.352	0.0232	0.0753	1.05	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.211	2.11	8.65	5.82	0.331	3.19	0.252	376	4.71e+03	0	Complete
4 T/P DeepONet10	**baseline**	3.80e-03	0.0682	0.0852	0.0645	2.64e-03	0.0271	4.83e-03	0.0327	0.023	0	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.154 ms**
5 ANN10 + T/P DeepONet10	**baseline**	4.76e-03	0.0786	0.0856	0.0623	1.90e-03	0.045	6.40e-03	0.0294	0.0218	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.0707	2.01	1.01	4.91	0.24	1.16	0.136	49	7.86	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	8.28e-03	0.0376	0.0669	0.036	0.03	0.288	0.0133	0.265	0.371	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.61	19	1.17	11.1	0.915	0.425	1	332	2.39e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	3.84e-03	0.0728	0.087	0.0721	0.0256	0.213	7.18e-03	0.213	0.325	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.546	12.3	1.25	6.65	0.295	5.28	0.827	3.29	2.89	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 64

T_0 = 1045.3 K P_0 = 4.301 atm ϕ_0 = 0.595 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 64 | NRMSE (util.py)

T0 = 1045.3 K P0 = 4.301 atm phi0 = 0.595 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.284	0.791	1.02	1.04	0.509	11.4	0.523	1.02	1.01	0	Complete
1 ANN10	**soft10k**	0.323	0.9	15.3	8.44	0.551	8.46	0.608	86.6	57.5	0	Complete
2 P-only DeepONet10	**baseline**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
2 P-only DeepONet10	**soft10k**	0.312	0.876	1	1	0.865	1	0.697	451	970	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.29	0.823	15	10.9	0.507	7.59	0.562	91.6	5.43	0	Complete
4 T/P DeepONet10	**baseline**	0.346	0.949	6.56	1.71	0.539	3.79	0.654	1.21	0.992	0	Complete
4 T/P DeepONet10	**soft10k**	0.381	0.935	9.45	1.06	0.491	1.35	0.694	1.02	0.991	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	0.331	0.919	4.92	1.11	0.528	4.59	0.625	1.09	0.99	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.365	0.974	11	9.5	0.541	2.07	0.712	46.6	1.97	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.399	0.98	11.9	1.12	0.513	1.44	0.728	1.81	1.61	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.542	1.37	1	1	0.733	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.379	0.973	7.91	5.32	0.535	2.14	0.715	60.2	2.96	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 68

T_0 = 1086.3 K P_0 = 3.702 atm ϕ_0 = 1.168 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

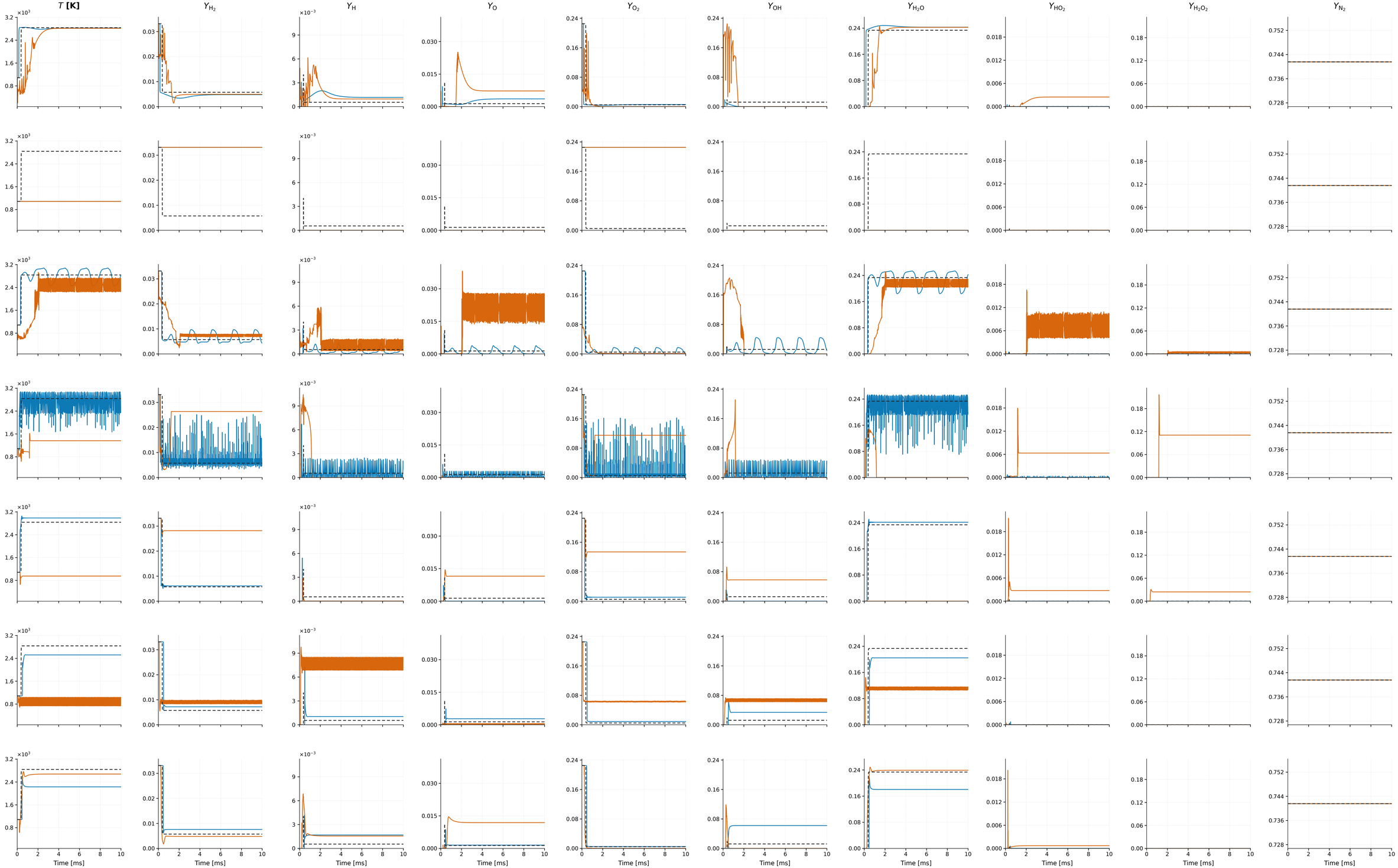
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 68 | NRMSE (util.py)

T0 = 1086.3 K P0 = 3.702 atm phi0 = 1.168 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.092	0.499	1.5	1.33	0.738	0.979	0.157	1.47	1.33	0	Complete
1 ANN10	**soft10k**	0.21	0.484	2.4	5.36	0.82	4.91	0.242	91.8	16.8	0	Complete
2 P-only DeepONet10	**baseline**	0.615	3.15	1	1	4.94	1	1	0.986	0.989	0	Complete
2 P-only DeepONet10	**soft10k**	0.615	3.15	1	0.999	4.94	1	1	1.41	4.16	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	0.0874	0.309	0.756	0.913	0.368	1.32	0.126	1.17	1.26	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.29	0.532	2.38	14.1	0.71	5.79	0.312	278	436	0	Complete
4 T/P DeepONet10	**baseline**	0.124	0.51	1.4	0.786	0.778	1.36	0.174	2.17	1.35	0	Complete
4 T/P DeepONet10	**soft10k**	0.53	2.31	4.6	1	2.39	2.59	0.971	253	1.23e+04	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	0.0776	0.325	1.01	0.993	0.528	1.02	0.102	1.28	1.09	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.662	2.58	1.01	7.32	2.89	3.78	1	118	2.82e+03	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.132	0.389	0.981	1.04	0.557	1.8	0.167	1.73	1.92	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.688	0.614	13.2	0.704	1.44	4.55	0.531	1.21	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.219	0.369	2.06	0.265	0.472	4.04	0.245	1.75	1.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.0665	0.326	2.4	7.57	0.48	1.38	0.0599	34.2	3.36	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 69

T_0 = 935.0 K P_0 = 2.045 atm ϕ_0 = 0.523 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

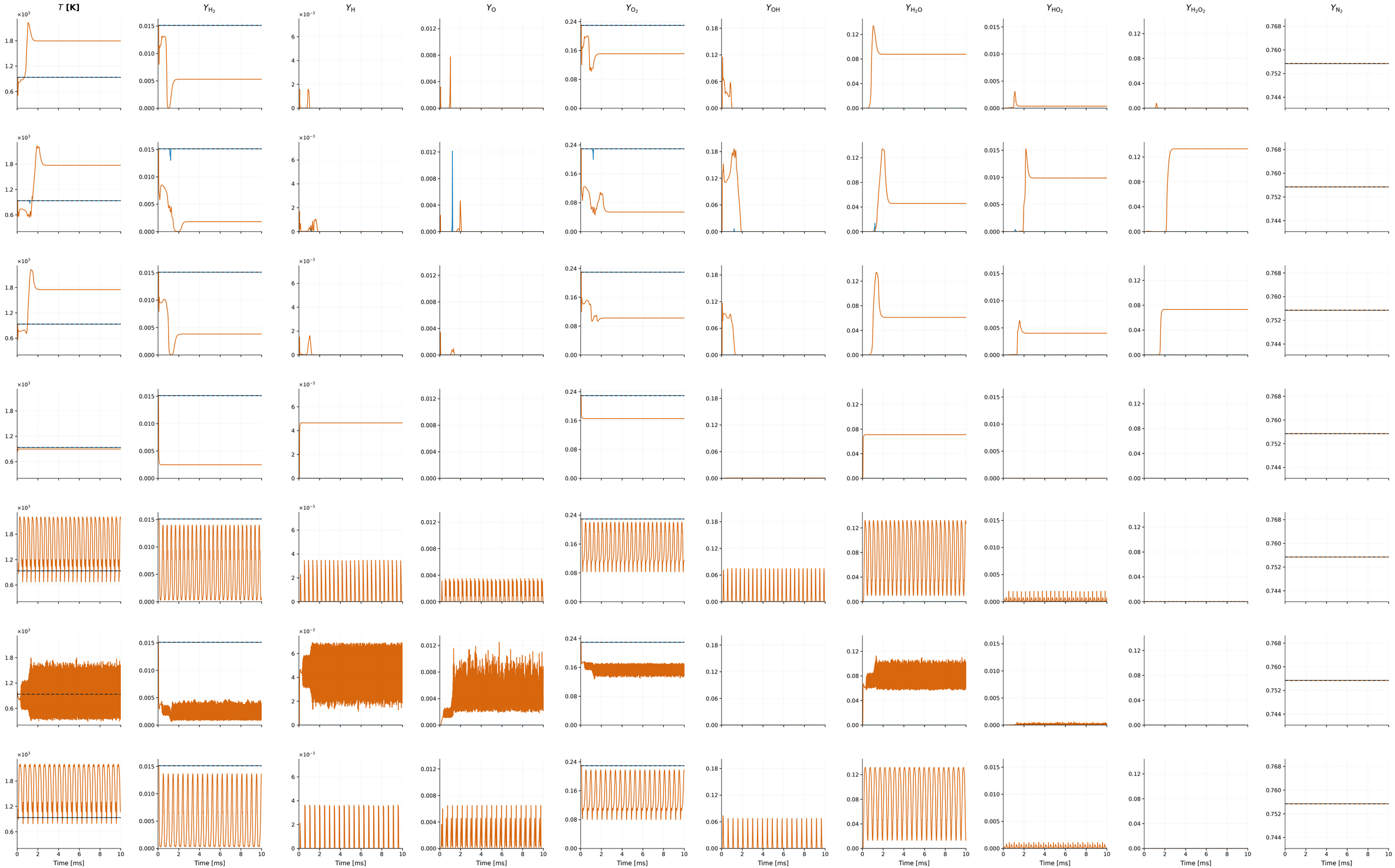
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 69 | NRMSE (util.py)

T0 = 935.0 K P0 = 2.045 atm phi0 = 0.523 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	9.88e-07	1.28e-06	1	185	1.53e-06	1	1	0.967	0.997	0	Complete
1 ANN10	**soft10k**	0.896	0.644	5.90e+07	4.44e+07	0.343	4.06e+09	1.28e+06	2.22e+03	8.1e+03	0	Complete
2 P-only DeepONet10	**baseline**	2.83e-03	8.45e-03	9.87e+06	5.45e+07	6.11e-03	7.94e+07	1.15e+04	139	52.9	0	Complete
2 P-only DeepONet10	**soft10k**	0.852	0.854	6.02e+07	4.02e+07	0.73	1.60e+10	7.59e+05	3.97e+04	1.33e+06	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	9.70e-07	1.29e-06	1	0.492	1.55e-06	0.998	0.999	0.988	0.999	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.85	0.735	7.06e+07	1.57e+07	0.538	7.93e+09	9.39e+05	1.69e+04	7.63e+05	0	Complete
4 T/P DeepONet10	**baseline**	9.45e-07	1.27e-06	1	2.28	1.55e-06	0.952	0.963	0.997	1	0	Complete
4 T/P DeepONet10	**soft10k**	0.0397	0.832	1.48e+09	1.25e+05	0.276	1.91e+08	1.06e+06	59.3	1	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	3.86e-04	5.04e-04	1	13	5.70e-04	0.999	8.88	94.2	1.33e+03	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.872	0.714	3.22e+08	1.13e+08	0.393	6.85e+09	1.31e+06	1.85e+03	2.89e+03	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	2.78e-05	2.32e-05	87.3	109	1.84e-05	399	35.7	7.04	8.79	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.474	0.857	1.49e+09	5.38e+08	0.322	6.67e+06	1.17e+06	476	0.977	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	9.40e-06	8.94e-06	18.9	16.9	8.37e-06	52.1	11.3	4.03	5.05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.934	0.757	3.44e+08	2.12e+08	0.411	5.37e+09	1.41e+06	1.49e+03	1.26e+03	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 73

T_0 = 1106.0 K P_0 = 1.911 atm ϕ_0 = 1.915 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

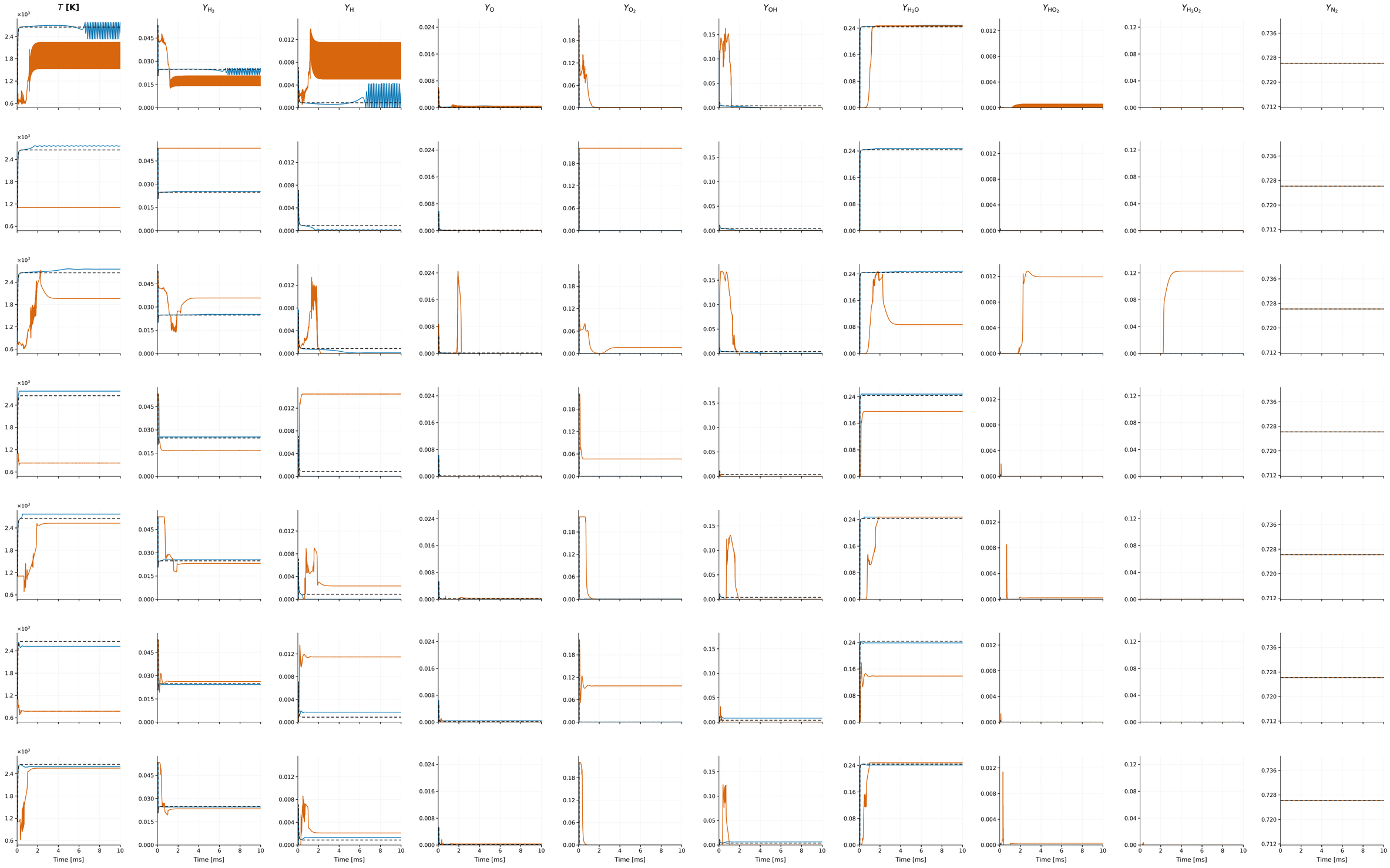
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 73 | NRMSE (util.py)

T0 = 1106.0 K P0 = 1.911 atm phi0 = 1.915 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.0381	0.0487	1.14	0.997	0.157	0.82	0.0187	0.373	6.14	0	Complete
1 ANN10	**soft10k**	0.384	0.371	8.39	1.47	1.57	10.4	0.302	36.7	176	0	Complete
2 P-only DeepONet10	**baseline**	0.0376	0.025	0.757	0.734	0.154	0.77	0.019	0.37	0.648	0	Complete
2 P-only DeepONet10	**soft10k**	0.582	1.13	1	1	12.3	1	1	1	1	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	0.03	0.0143	0.592	0.58	7.95e-03	0.633	0.0124	0.0794	0.188	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.344	0.437	2.8	14.4	1.41	12.3	0.636	1.04e+03	2.27e+05	0	Complete
4 T/P DeepONet10	**baseline**	0.0466	0.0258	0.956	0.772	0.0161	0.986	0.0171	0.459	6.48	0	Complete
4 T/P DeepONet10	**soft10k**	0.681	0.331	14.7	0.991	2.85	0.99	0.218	8.18	0.969	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	0.0444	0.0276	0.93	0.673	0.0709	0.961	0.0159	0.422	2.4	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.229	0.305	2.55	1	3.17	7.79	0.299	28.8	167	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.0497	0.0333	0.96	1.13	0.242	1.05	0.0279	0.634	0.879	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.706	0.107	11.4	0.899	5.48	1.01	0.437	6.21	1.13	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.0248	0.0245	0.472	0.561	0.21	0.58	0.0195	0.528	0.672	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.159	0.204	2.05	0.985	2.09	4.76	0.196	41	414	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 74

T_0 = 1222.9 K P_0 = 4.971 atm ϕ_0 = 1.766 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

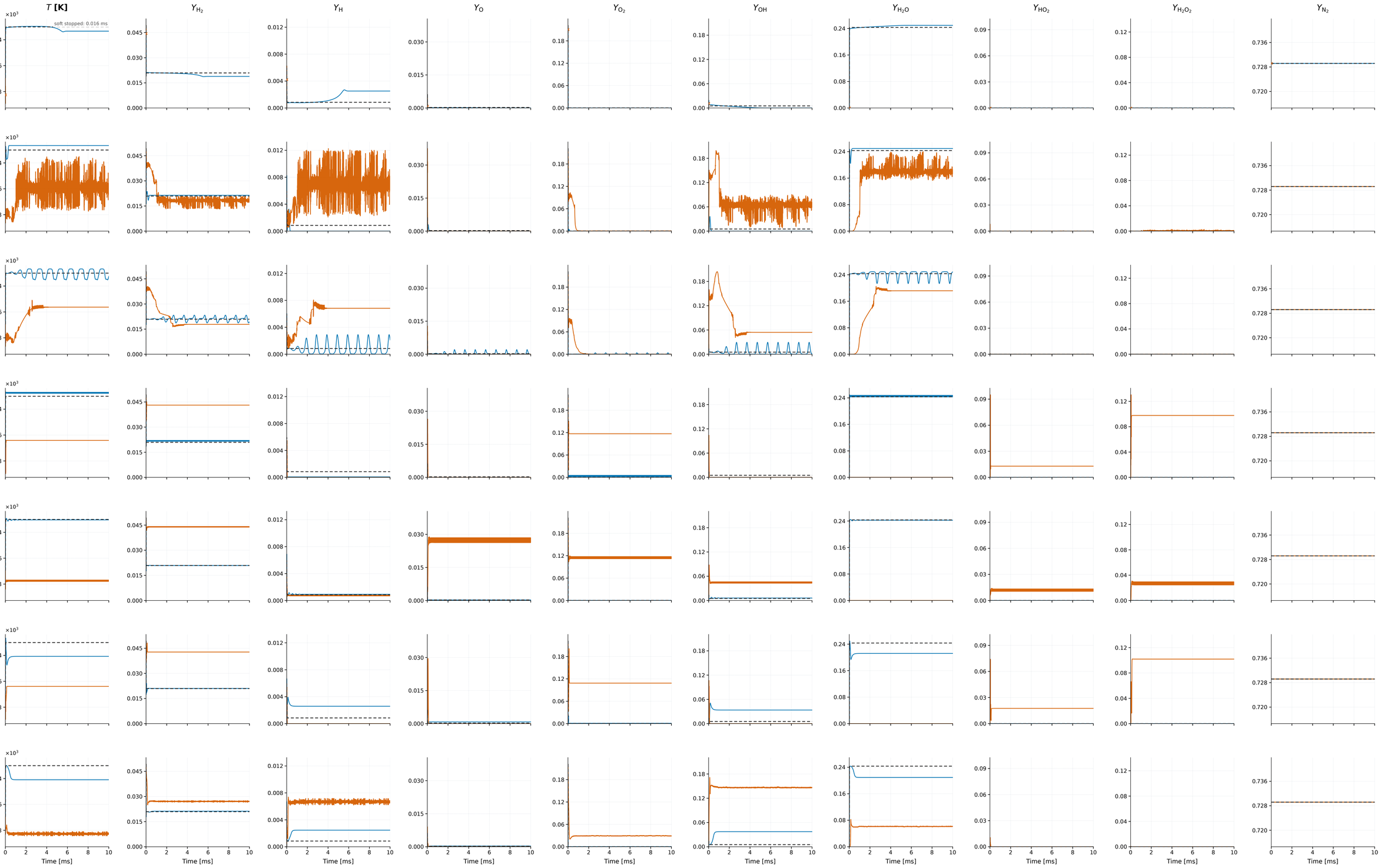
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 74 | NRMSE (util.py)

T0 = 1222.9 K P0 = 4.971 atm phi0 = 1.766 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.0331	0.0696	1.38	0.728	0.0408	0.804	0.0201	0.211	1.99	0	Complete
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.016 ms**
2 P-only DeepONet10	**baseline**	0.0501	0.0284	1.01	1.17	0.177	1.1	0.0287	0.692	1.41	0	Complete
2 P-only DeepONet10	**soft10k**	0.469	0.272	7.4	3.92	2.78	14.3	0.386	13.6	734	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	0.045	0.0562	1.06	2.06	0.212	1.67	0.0447	0.662	3.9	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.447	0.277	6.44	1.57	2.51	15.6	0.411	1.89	78.6	0	Complete
4 T/P DeepONet10	**baseline**	0.04	0.0532	0.97	1.09	0.72	1	0.0224	1.15	9.51	0	Complete
4 T/P DeepONet10	**soft10k**	0.488	1.06	1	3.99	14.1	1.21	1	1.53e+03	1.34e+05	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	8.40e-03	0.0292	0.178	0.524	0.521	0.199	0.0177	1.07	0.981	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.676	1.09	0.163	99.1	13.9	7.54	1	1.41e+03	3.82e+04	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.155	0.0323	2.06	2	0.572	5.48	0.131	1.5	5.07	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.486	1.04	0.995	7	13.2	1.38	1	2.04e+03	1.39e+05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.152	0.0322	1.86	0.606	0.517	5.99	0.138	1.4	6.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.751	0.308	6.79	1.02	3.79	27	0.751	52.2	45.6	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 76

T_0 = 1523.6 K P_0 = 3.626 atm ϕ_0 = 1.349 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

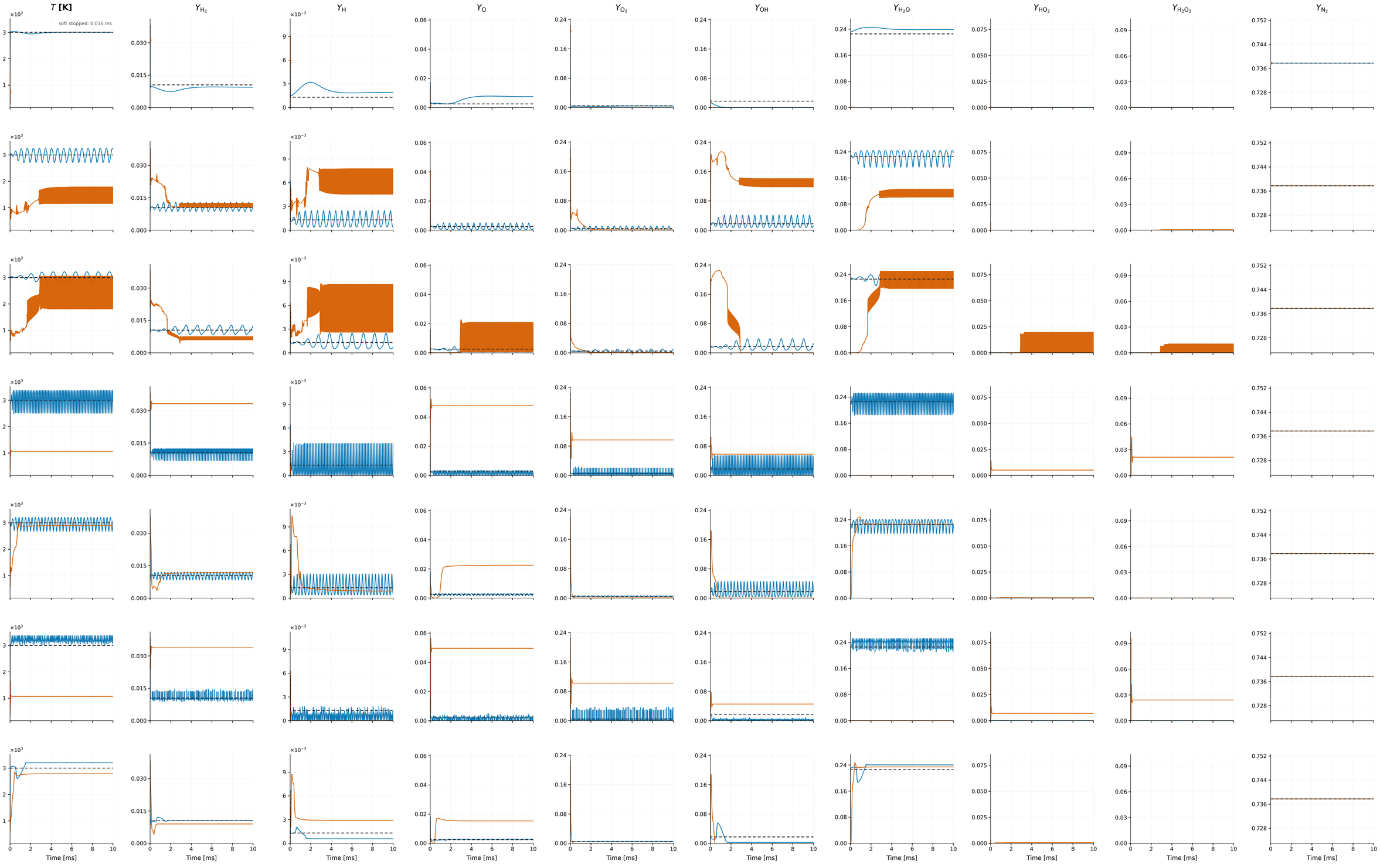
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 76 | NRMSE (util.py)

T0 = 1523.6 K P0 = 3.626 atm phi0 = 1.349 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	7.28e-03	0.157	0.733	1.66	0.164	0.967	0.0661	3.08	10.3	0	Complete
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.016 ms**
2 P-only DeepONet10	**baseline**	0.0606	0.124	0.537	0.631	0.585	0.637	0.075	1.43	3.56	0	Complete
2 P-only DeepONet10	**soft10k**	0.567	0.463	3.76	1.08	2.36	7.31	0.609	19.4	427	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	0.0447	0.119	0.469	0.535	0.487	0.537	0.062	1.43	4.32	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.38	0.524	3.64	4.15	1.73	4.74	0.406	1.59e+03	6.13e+03	0	Complete
4 T/P DeepONet10	**baseline**	0.101	0.138	1.02	0.564	0.519	1.13	0.102	4.47	2.83	0	Complete
4 T/P DeepONet10	**soft10k**	0.646	2.16	0.995	18	14	2.33	1	783	2.67e+04	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	0.0593	0.113	0.713	0.206	0.165	0.838	0.067	0.92	1.56	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.12	0.253	1.38	7.42	1.72	1.42	0.116	44	52.4	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.0794	0.0678	0.632	0.485	0.728	0.848	0.0779	2.08	2.75	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.643	2.2	1	18.7	14.9	1.61	1	1.13e+03	3.05e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.0672	0.0325	0.52	0.132	0.106	0.894	0.0693	0.916	1.18	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.119	0.229	1.5	5.03	1.75	1.37	0.113	108	56.1	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 78

T_0 = 962.3 K P_0 = 2.443 atm ϕ_0 = 1.140 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

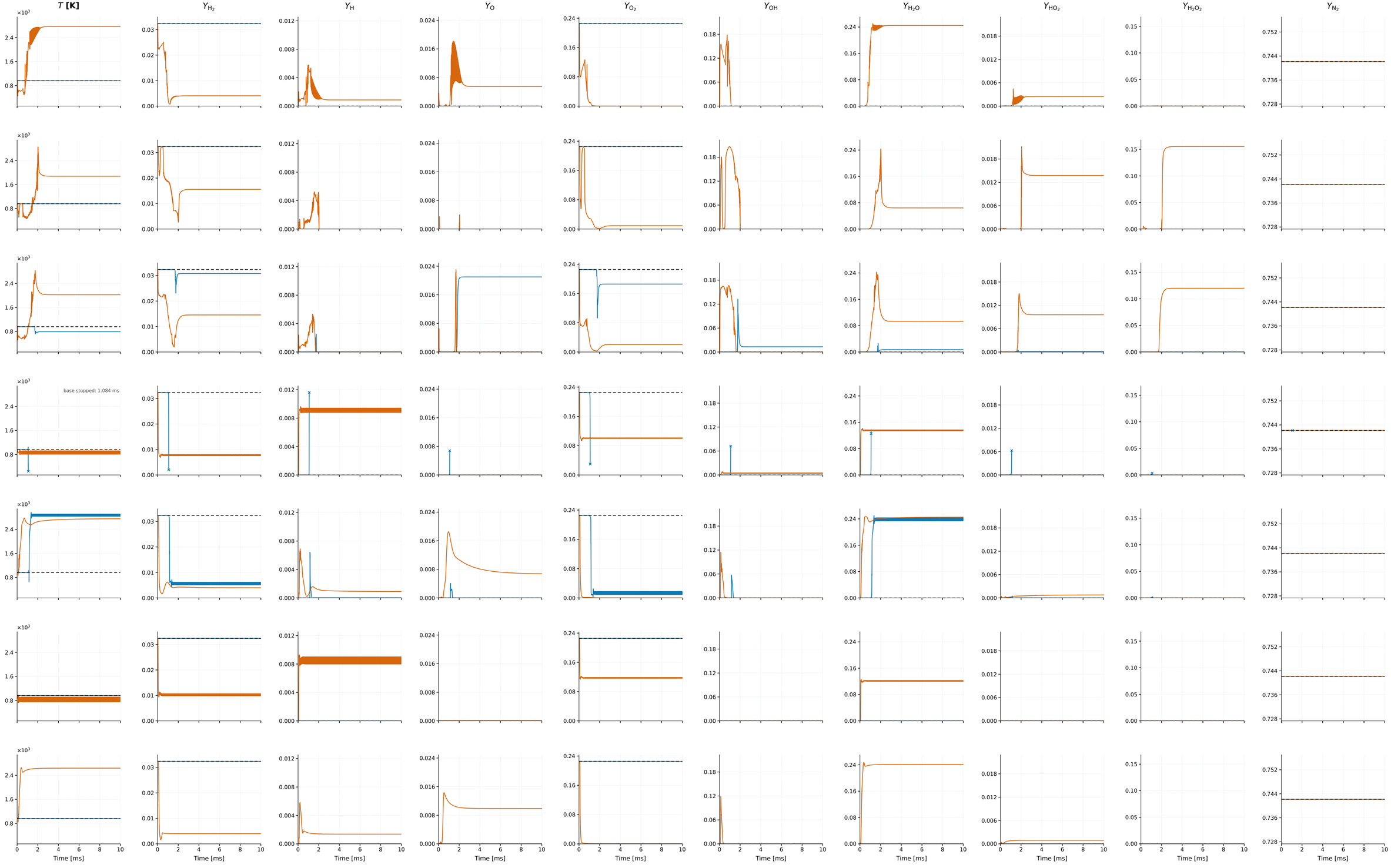
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 78 | NRMSE (util.py)

T0 = 962.3 K P0 = 2.443 atm phi0 = 1.140 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	4.72e-05	2.63e-05	3.24	4.29e+03	6.52e-05	1	1	1.07	2.06	0	Complete
1 ANN10	**soft10k**	1.75	0.844	7.46e+06	2.11e+07	0.97	2.88e+08	6.18e+04	480	34.4	0	Complete
2 P-only DeepONet10	**baseline**	3.54e-05	1.95e-05	1	0.876	4.00e-05	0.999	0.869	0.848	0.718	0	Complete
2 P-only DeepONet10	**soft10k**	0.881	0.522	5.79e+06	6.43e+05	0.93	4.64e+08	1.83e+04	2.63e+03	2.96e+04	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	0.16	0.0496	8.47e+05	6.63e+07	0.166	1.05e+08	1.77e+03	16.5	4.06	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	1.06	0.561	5.41e+06	6.36e+06	0.894	3.85e+08	2.64e+04	1.89e+03	2.31e+04	0	Complete
4 T/P DeepONet10	**baseline**	—	—	—	—	—	—	—	—	—	—	**Stop 1.084 ms**
4 T/P DeepONet10	**soft10k**	0.112	0.756	4.75e+07	3.2e+03	0.552	3.23e+07	3.61e+04	2.57	1	0	Complete
5 ANN10 + T/P DeepONet10	**baseline**	1.86	0.777	1.64e+06	1.13e+06	0.886	3.31e+07	5.98e+04	8.67	37.4	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	1.79	0.868	7.14e+06	2.99e+07	0.992	8.47e+07	6.41e+04	137	11.4	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	4.54e-05	2.59e-05	1	0.884	5.42e-05	0.997	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.156	0.681	4.44e+07	8.67e+04	0.477	7.00e+04	3.23e+04	0.726	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	4.54e-05	2.58e-05	1	0.819	5.42e-05	0.995	0.999	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	1.71	0.872	8.09e+06	3.49e+07	0.991	7.41e+07	6.36e+04	186	7.66	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Soft constraint: Cantera vs NN | Test IC 88

T_0 = 1502.1 K P_0 = 4.980 atm ϕ_0 = 0.859 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10)

1 ANN10

2 P-only DeepONet10

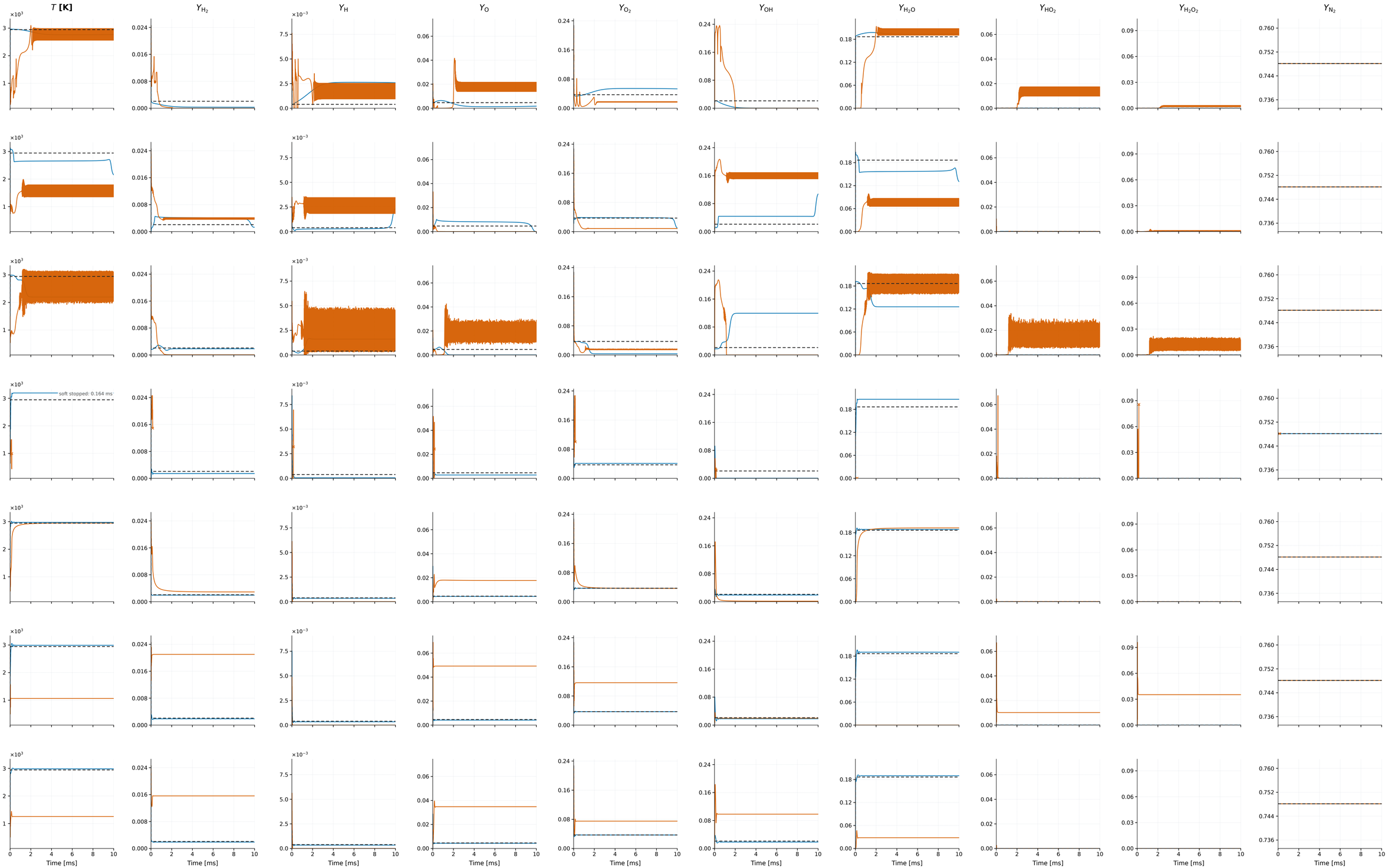
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
All networks have10000 updates. Mixed teams use the corresponding baseline or soft ANN10 + DeepONet10 at the same current state. Soft training adds mu10 physical element L1; inference rules are unchanged.

Cantera vs NN | Test IC 88 | NRMSE (util.py)

T0 = 1502.1 K P0 = 4.980 atm phi0 = 0.859 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**baseline**	0.0536	0.764	5.02	0.609	0.392	0.914	0.0409	1.27	4.48	0	Complete
1 ANN10	**soft10k**	0.224	1.3	4.59	2.8	0.599	2.9	0.282	449	1.05e+03	0	Complete
2 P-only DeepONet10	**baseline**	0.101	0.92	0.803	0.75	0.101	1.19	0.157	5.76	17.3	0	Complete
2 P-only DeepONet10	**soft10k**	0.489	1.35	6.09	1	0.775	6.68	0.625	6.29	356	0	Complete
3 ANN10 + P-only DeepONet10	**baseline**	0.234	0.143	2.9	0.928	0.832	4.28	0.3	1.4	7.8	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.249	1.3	6.19	3.35	0.612	2.55	0.27	558	5.32e+03	0	Complete
4 T/P DeepONet10	**baseline**	0.0871	0.306	1.15	0.406	0.106	0.998	0.107	2.11	2.76	0	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.164 ms**
5 ANN10 + T/P DeepONet10	**baseline**	0.0113	0.0567	0.186	0.0697	0.0111	0.105	0.0136	0.111	0.106	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.0755	1.02	1.11	2.8	0.307	1.05	0.122	3.37	7.39	0	Complete
6 T/P/ϕ DeepONet10	**baseline**	0.024	0.107	0.479	0.124	0.0122	0.217	0.0275	0.295	0.232	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.639	8.9	1.05	9.61	2.12	0.0811	1	376	1.67e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.0144	0.076	0.19	0.0787	0.012	0.145	0.0184	0.122	0.129	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.573	6.37	1.08	6.49	1.03	3.68	0.85	2.5	1.25	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.